

Lesson 9: Introduction to the Bootstrap

BSTA 551: Statistical Inference

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Lesson 9: The Bootstrap

Review: Where We Left Off

Lesson 8: One-Sample t-Intervals and Population Proportions

- t-intervals when σ is unknown: $\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
- Normality assumption for t-intervals
- Confidence intervals for proportions (Wald, Score, Agresti-Coull)

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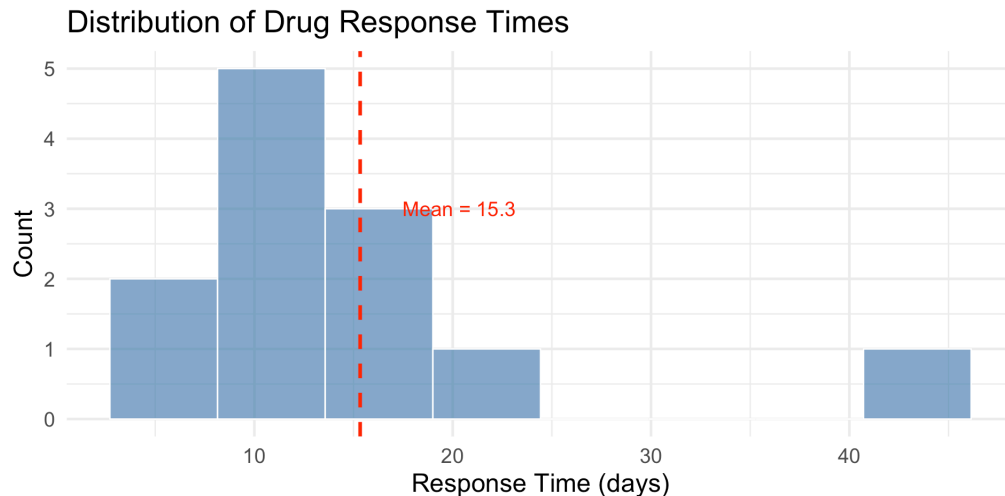
Today's Goals:

1. Understand the bootstrap principle and its theoretical foundation
2. Learn why resampling **with replacement** is essential
3. Implement the basic bootstrap algorithm in R
4. Explore asymptotic properties of bootstrap estimates
5. Understand when bootstrap methods fail

Motivating Example: Drug Response Times

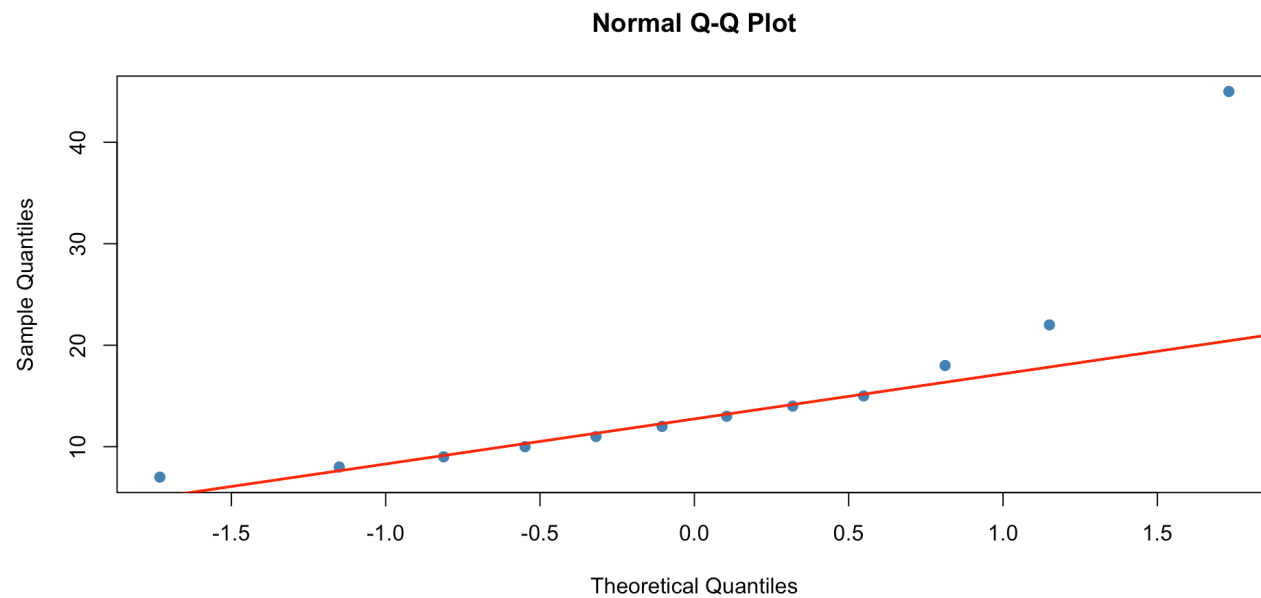
In a clinical trial, 12 patients' response times (days) to a new medication:

```
1 response_times <- c(8, 12, 15, 7, 22, 45, 11, 9, 14, 18, 10, 13)
2
3 tibble(x = response_times) |>
4   ggplot(aes(x = x)) +
5   geom_histogram(bins = 8, fill = "steelblue", color = "white", alpha = 0.7) +
6   geom_vline(xintercept = mean(response_times), color = "red", linewidth = 1.2, linetype =
7   labs(x = "Response Time (days)", y = "Count",
8         title = "Distribution of Drug Response Times") +
9   annotate("text", x = mean(response_times) + 5, y = 3,
10          label = paste0("Mean = ", round(mean(response_times), 1)), color = "red", size =
```



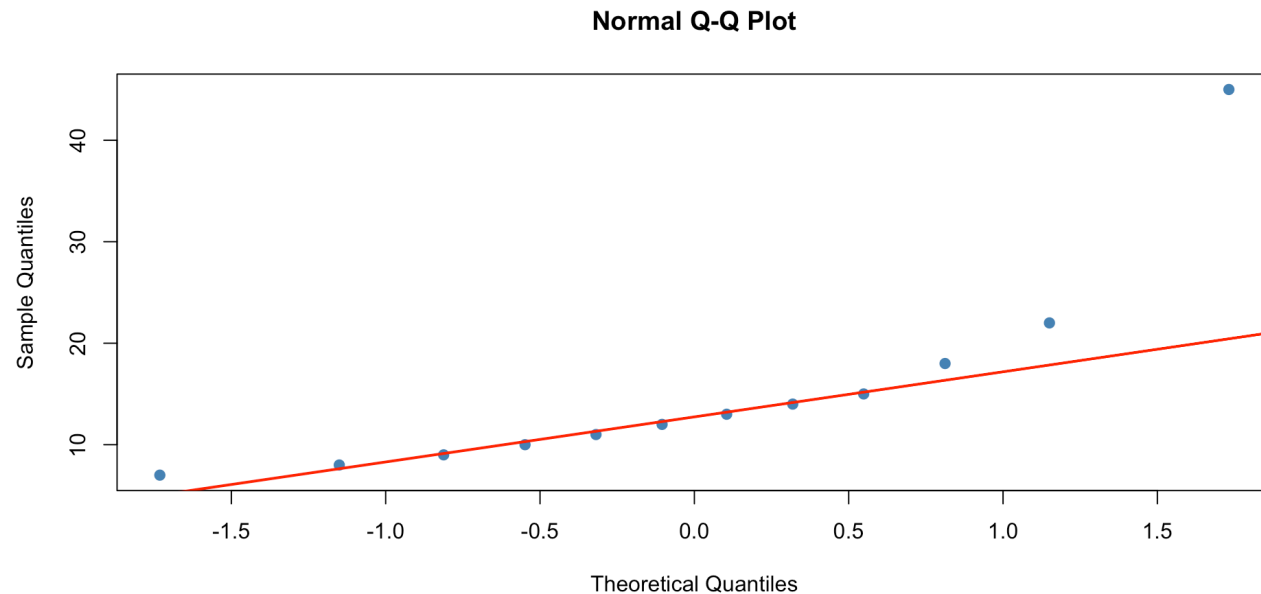
The Problem with Small Samples

► Code



The Problem with Small Samples

► Code



Issues:

- Small sample ($n = 12$)
- Evidence of right skew (outlier at 45 days)
- Normal-based t-interval may not have correct coverage

The Key Question

How can we construct confidence intervals when:

1. The population distribution is **unknown or non-normal**?
2. The sample size is **too small** for CLT to apply reliably?
3. We want to estimate parameters **other than the mean** (median, trimmed mean, ratio)?

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The Bootstrap Solution

Use the data itself to approximate the sampling distribution of our statistic!

The Bootstrap Idea

What is the Bootstrap?

! Definition (Devore, Berk, Carlton 8.5)

The **bootstrap** is a resampling method that uses the observed sample to approximate the sampling distribution of a statistic.

The method was developed by **Bradley Efron** in the late 1970s and substitutes heavy computation for theoretical derivation.

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Key insight: The sample is our best estimate of the population. If we resample *from the sample*, we can learn about sampling variability.

Traditional vs. Bootstrap Approach

Traditional Inference:

1. Assume population distribution (e.g., Normal)
2. Derive sampling distribution theoretically
3. Use theoretical distribution to construct intervals

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Bootstrap Inference:

1. Use the sample as a proxy for the population
2. Generate sampling distribution empirically via resampling
3. Use empirical distribution for inference

The Plug-In Principle (Chihara & Hesterberg 5.2)

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To estimate a parameter (a quantity that describes the **population**), use the statistic that is the corresponding quantity for the **sample**.

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Examples:

Population Parameter	Sample Statistic
Mean μ	$\bar{x}$
Standard deviation σ	s
Median $\tilde{\mu}$	Sample median
Standard error $\sigma/\sqrt{n}$	$s/\sqrt{n}$

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Standard error $\sigma/\sqrt{n}$	$s/\sqrt{n}$

The bootstrap extends this: Use the **entire sample** as an estimate of the **entire population**.

The Empirical Distribution Function

! Definition: Empirical CDF

Given a sample $x_1, x_2, \dots, x_n$, the **empirical cumulative distribution function** is:

$$\hat{F}_n(x) = \frac{1}{n} \#\{x_i \leq x\} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(x_i \leq x)$$

This assigns probability $1/n$ to each observed data point.

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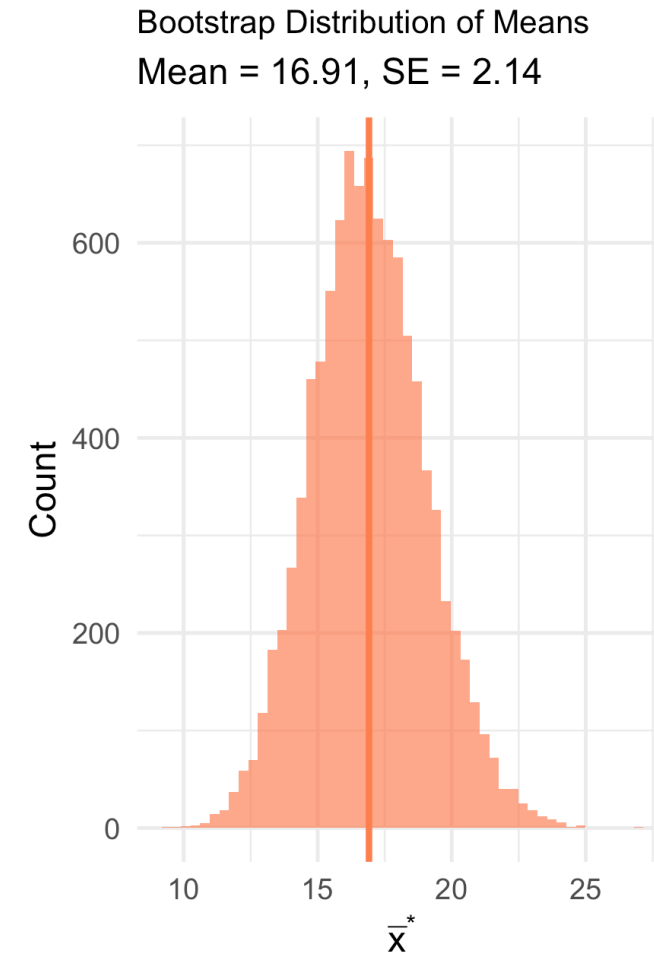
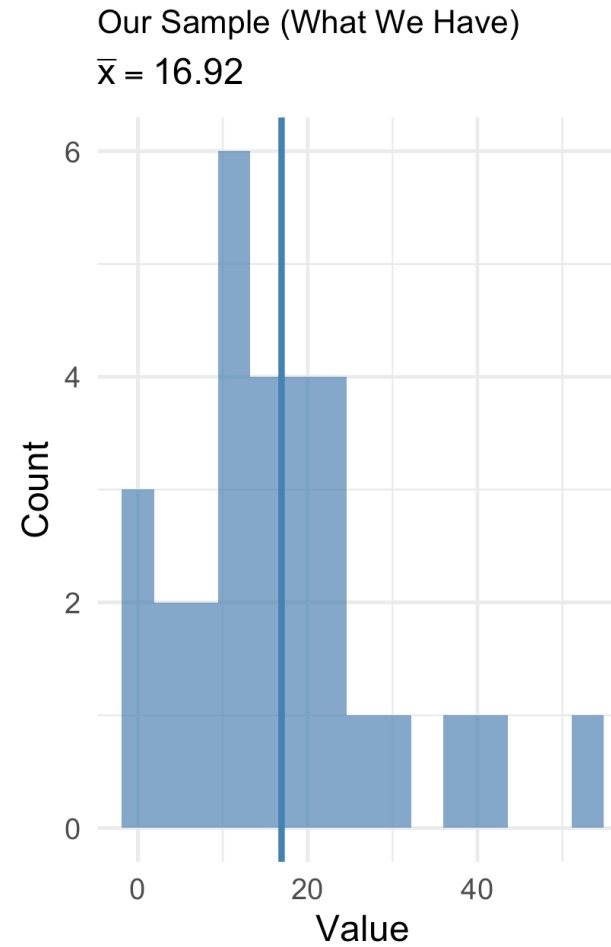
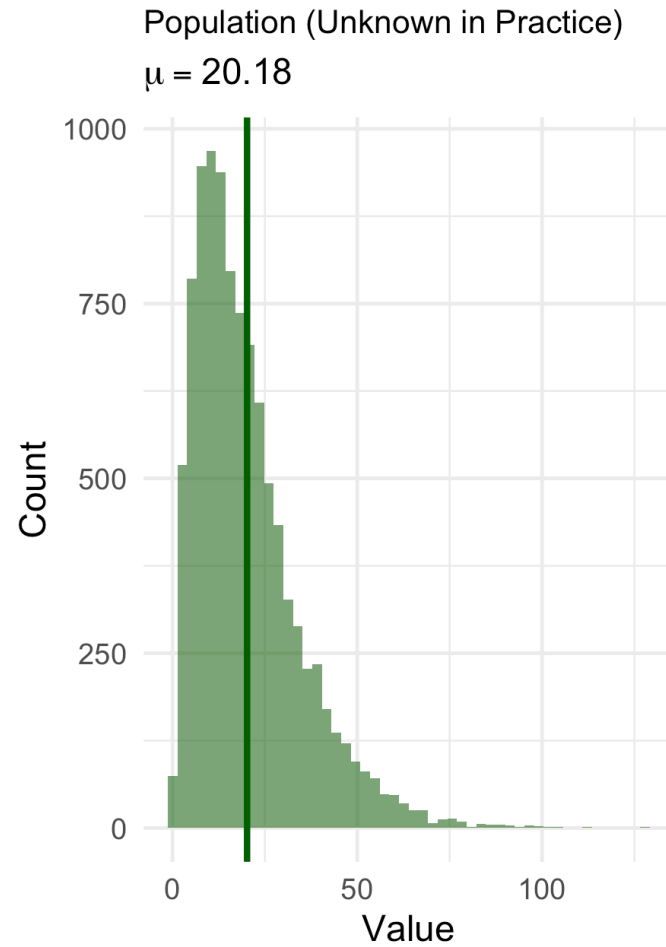
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Key properties of $\hat{F}_n$:

- $E_{\hat{F}}[X] = \bar{x}$ (mean under empirical distribution = sample mean)
- $\text{Var}_{\hat{F}}[X] = \frac{1}{n} \sum (x_i - \bar{x})^2$ (slightly different from s^2)

Visualizing the Bootstrap Idea

► Code



Sampling Distribution vs. Bootstrap Distribution

Sampling Distribution:

- Draw **many samples** from the **population**
- Calculate statistic for each
- Distribution of these statistics

In practice: We can't do this (we only have one sample!)

Bootstrap Distribution:

- Draw **many resamples** from the **sample**
- Calculate statistic for each
- Distribution of these statistics

In practice: We can do this computationally!

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Key Point

The bootstrap distribution **approximates** the sampling distribution, allowing us to estimate standard errors and construct confidence intervals.

Why Sampling WITH Replacement?

The Critical Question

Why must we sample **with replacement** rather than **without replacement** (subsampling)?

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Without Replacement

If we sample **without replacement** from our sample of size n , we would get:

- A subsample of size $m < n$: Less information, different properties
- A sample of size n : **Exactly the same sample every time!**

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Why must we sample **with replacement** rather than **without replacement** (subsampling)?

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Sampling with replacement allows:

1. Observations to appear multiple times (or not at all)
2. Variability in the bootstrap samples
3. Approximation of the sampling process from the population

Mathematical Justification

! Why Replacement Matters

When sampling from a population of size N , we sample with replacement because:

1. The population is treated as **infinite** (or very large relative to n)
2. Each draw is **independent** of previous draws
3. Each observation has the **same probability** of selection

The empirical distribution $\hat{F}_n$ is a **discrete** distribution with n equally likely values. Sampling from it **with replacement** mimics sampling from the true population.

Demonstrating: With vs. Without Replacement

► Code

With replacement (5 examples):

► Code

Sample 1 : 5 9 11 5 11 -> mean = 8.2

Sample 2 : 7 3 3 9 3 -> mean = 5

Sample 3 : 5 9 9 5 9 -> mean = 7.4

Sample 4 : 7 5 11 5 11 -> mean = 7.8

Sample 5 : 3 7 11 7 7 -> mean = 7

► Code

Without replacement:

► Code

Always get: 5 3 11 7 9

Number of Possible Bootstrap Samples

Counting Bootstrap Samples

For a sample of size n , the number of possible bootstrap samples (sampling n with replacement) is:

$$\binom{2n - 1}{n - 1} = \binom{2n - 1}{n}$$

This is the number of ways to choose n items from n categories with repetition allowed.

```
1 # Number of bootstrap samples for various n
2 tibble(n = c(5, 10, 15, 20, 30)) |>
3   mutate(
4     num_bootstrap_samples = choose(2*n - 1, n - 1),
5     formatted = format(num_bootstrap_samples, big.mark = ",", scientific = FALSE)
6   )
```

A tibble: 5 × 3

	n	num_bootstrap_samples	formatted
	<dbl>	<dbl>	<chr>
1	5	1.26e 2	"126"
2	10	9.24e 4	"92,378"
3	15	7.76e 7	"77,558,760"
4	20	6.89e10	"68,923,264,410"
5	30	5.91e16	"59,132,290,782,430,704"

Even for $n = 15$, there are over **77 million** possible bootstrap samples!

The Variance Connection

! Variance of Bootstrap Mean

For the sample mean, the **theoretical** (exhaustive) bootstrap standard error is:

$$SE_{boot}(\bar{X}^*) = \frac{\hat{\sigma}}{\sqrt{n}} = \sqrt{\frac{1}{n^2} \sum_{i=1}^n (x_i - \bar{x})^2}$$

where $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ is the population variance under $\hat{F}_n$.

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Note: This differs slightly from $s/\sqrt{n}$ by a factor of $\sqrt{(n-1)/n}$:

$$SE_{boot} = \sqrt{\frac{n-1}{n}} \cdot \frac{s}{\sqrt{n}}$$

This is why bootstrap SEs are slightly **narrower** than classical SEs.

Subsampling: An Alternative Approach

Subsampling (Without Replacement)

An alternative to bootstrap is **subsampling**: draw samples of size $m < n$ **without replacement**.

Properties:

- Requires choosing subsample size m
- Typically $m/n \rightarrow 0$ as $n \rightarrow \infty$
- Can work when bootstrap fails (heavy tails, dependent data)
- Generally less efficient than bootstrap

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When to use subsampling:

- Time series or dependent data
- Estimating extreme quantiles
- When bootstrap consistency fails

The Bootstrap Method

Basic Bootstrap Algorithm (Devore 8.5)

! Basic Bootstrap Method

Given a sample $x_1, x_2, \dots, x_n$ and a statistic $\hat{\theta}$:

1. **Resample:** Draw a sample of size n **with replacement** from $x_1, \dots, x_n$
2. **Compute:** Calculate $\hat{\theta}^*$ from this bootstrap sample
3. **Repeat:** Repeat steps 1-2 many times (typically $B = 1000$ to 10000)
4. **Analyze:** Use the distribution of $\hat{\theta}_1^*, \hat{\theta}_2^*, \dots, \hat{\theta}_B^*$ for inference

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Why sample with replacement?

Without replacement, we'd get the same sample every time!

Simple Example: Birth Weights (Chihara 5.1)

Consider just 3 birth weights: 3969, 3204, 2892 grams

```
1 small_sample <- c(3969, 3204, 2892)
2 cat("Original sample:", small_sample, "\n")
```

Original sample: 3969 3204 2892

```
1 cat("Sample mean:", mean(small_sample))
```

Sample mean: 3355

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Sample mean: 3355

All possible bootstrap samples (sampling with replacement):

► Code

```
# A tibble: 10 × 5
      x1     x2     x3 sample_id  mean
  <dbl> <dbl> <dbl>    <int> <dbl>
1  3969  3969  3969         1  3969
2  3969  3969  3204         2  3714
3  3969  3969  2892         3  3610
4  3969  3204  3969         4  3714
5  3969  3204  3204         5  3459
6  3969  3204  2892         6  3355
7  3969  2892  3969         7  3610
8  3969  2892  3204         8  3355
9  3969  2892  2892         9  3251
```

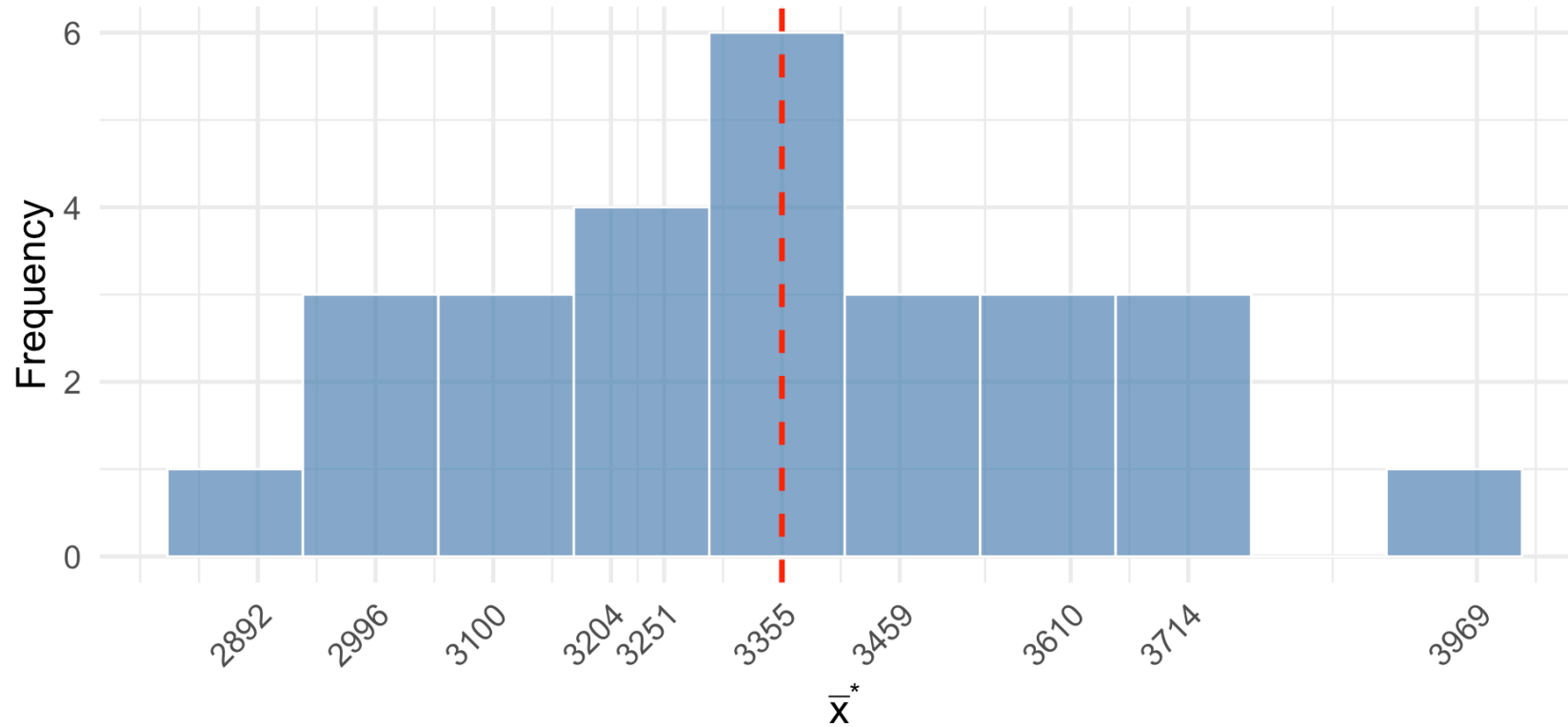


Complete Bootstrap Distribution (Small Example)

► Code

Bootstrap Distribution of Mean (All 27 Samples)

Original sample mean = 3355



Bootstrap standard error: 266.2 grams



Bootstrap in R: Step by Step

```
1 # Original response time data
2 response_times <- c(8, 12, 15, 7, 22, 45, 11, 9, 14, 18, 10, 13)
3 n <- length(response_times)
4 B <- 10000 # Number of bootstrap samples
5
6 # Storage for bootstrap means
7 boot_means <- numeric(B)
8
9 # Bootstrap loop
10 for (i in 1:B) {
11   # Step 1: Resample with replacement
12   boot_sample <- sample(response_times, size = n, replace = TRUE)
13
14   # Step 2: Calculate statistic
15   boot_means[i] <- mean(boot_sample)
16 }
17
18 # Results
19 cat("Original sample mean:", round(mean(response_times), 2), "\n")
```

Original sample mean: 15.33

```
1 cat("Bootstrap mean:", round(mean(boot_means), 2), "\n")
```

Bootstrap mean: 15.3

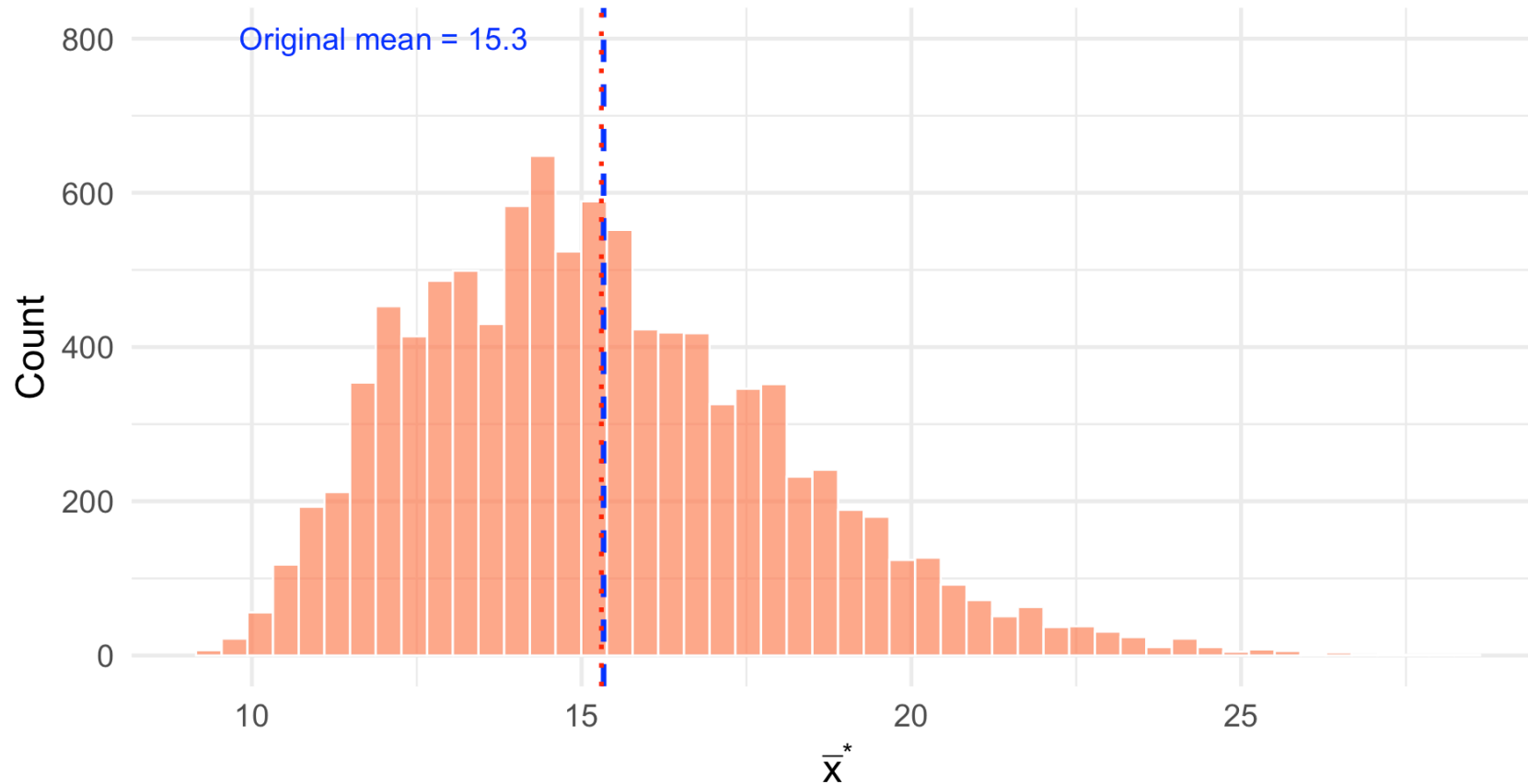


Bootstrap Distribution: Drug Response Times

► Code

Bootstrap Distribution of Mean Response Time

B = 10000 resamples; SE = 2.82 days



Using Tidyverse for Bootstrap

```
1 # More elegant approach using tidyverse
2 set.seed(551)
3 boot_results <- tibble(
4   replicate = 1:B
5 ) |>
6   mutate(
7     boot_sample = map(replicate, ~sample(response_times, n, replace = TRUE)),
8     boot_mean = map_dbl(boot_sample, mean),
9     boot_median = map_dbl(boot_sample, median),
10    boot_sd = map_dbl(boot_sample, sd))
11 boot_results |>
12   select(-boot_sample) |>
13   summarize(
14     across(starts_with("boot_"), list(mean = mean, se = sd))
15   ) |>
16   pivot_longer(everything(), names_to = "statistic", values_to = "value")
```

A tibble: 6 × 2

	statistic	value
	<chr>	<dbl>
1	boot_mean_mean	15.3
2	boot_mean_se	2.82
3	boot_median_mean	12.5
4	boot_median_se	1.70

Asymptotic Properties of Bootstrap

Bootstrap Consistency

! Bootstrap Consistency Theorem

Under regularity conditions, the bootstrap distribution consistently estimates the sampling distribution. Specifically:

Let $\hat{\theta}_n$ be an estimator of θ based on a sample of size n . If:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$$

Then under the bootstrap:

$$\sqrt{n}(\hat{\theta}_n^* - \hat{\theta}_n) \xrightarrow{d} N(0, \sigma^2)$$

where convergence is in probability.

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where convergence is in probability.

In plain English: The bootstrap distribution mimics the true sampling distribution as $n \rightarrow \infty$.

What Bootstrap Estimates Well

Bootstrap Accurately Estimates (Chihara 5.2.2)

1. **Spread:** The spread (standard error) of the sampling distribution
2. **Shape:** The shape (including skewness) of the sampling distribution
3. **Bias:** The bias of the estimator

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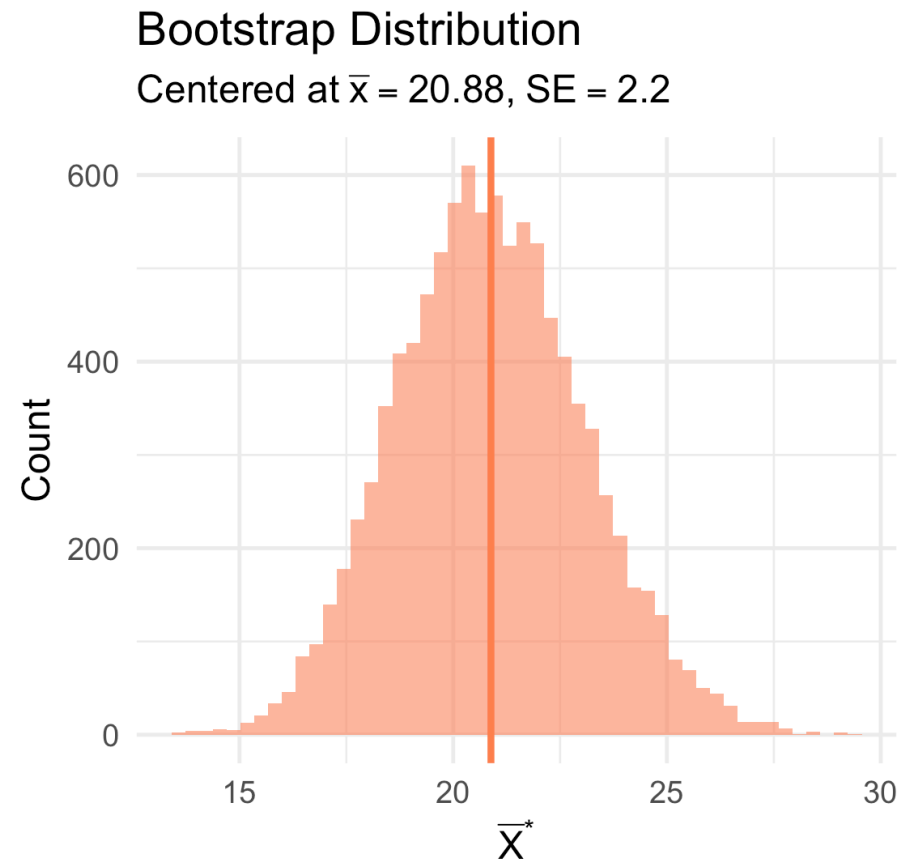
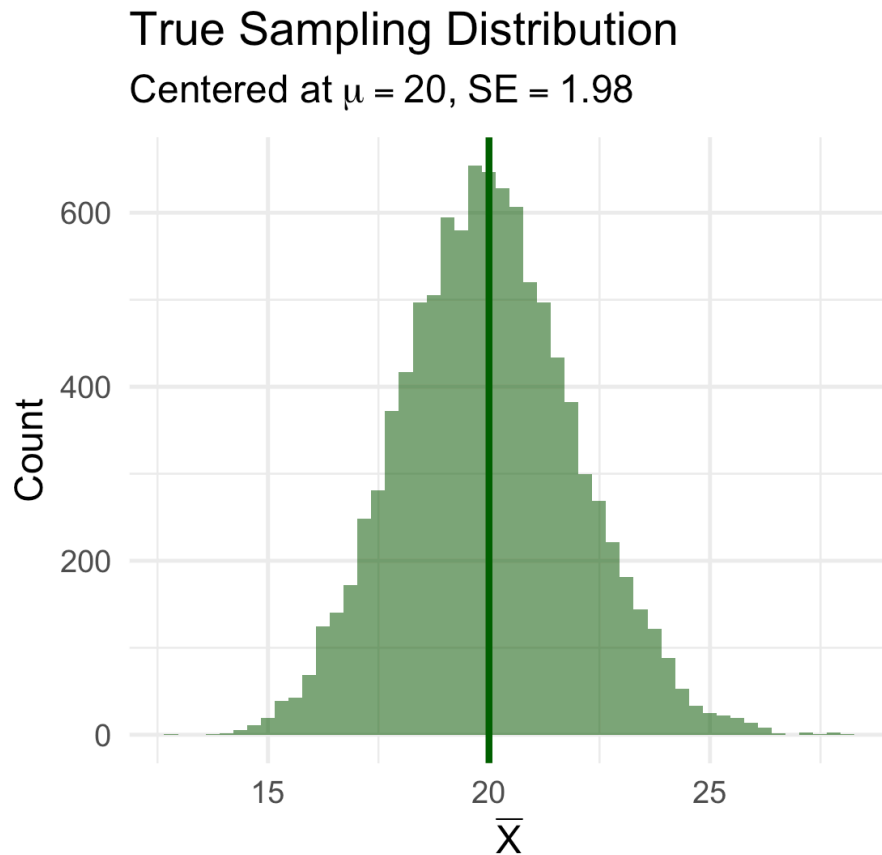
Bootstrap Does NOT Estimate

Center: The bootstrap distribution is centered at $\hat{\theta}$ (the sample statistic), NOT at θ (the true parameter).

This means bootstrap **cannot improve point estimates!**

Visualizing What Bootstrap Captures

► Code



Key observation: Same spread and shape, but different centers!

Comparison Table (Chihara Table 5.2)

Quantity	Mean	Standard Deviation
Population	μ	σ
Sampling distribution of $\bar{X}$	μ	$\sigma/\sqrt{n}$
Sample	$\bar{x}$	s
Bootstrap distribution	$\approx \bar{x}$	$\approx s \cdot \sqrt{((n-1)/n)/\sqrt{n}}$

Comparison Table (Chihara Table 5.2)

Quantity	Mean	Standard Deviation
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Sample	$\bar{x}$	s
Bootstrap distribution	$\approx \bar{x}$	$\approx s \cdot \sqrt{(n-1)/n} / \sqrt{n}$

Key insights:

- Bootstrap mean $\approx$ sample mean (not population mean)
- Bootstrap SE $\approx$ sampling distribution SE
- Bootstrap shape $\approx$ sampling distribution shape

Rate of Convergence

Edgeworth Expansion Refinement

For the sample mean, classical confidence intervals have coverage error of order $O(n^{-1/2})$.

Bootstrap percentile intervals: Also $O(n^{-1/2})$ — same as classical

Bootstrap t intervals: $O(n^{-1})$ — **one order more accurate!**

This is because the bootstrap t automatically corrects for skewness.

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Practical implication: Bootstrap t intervals are preferred for skewed distributions, especially with moderate sample sizes.

When Bootstrap Fails

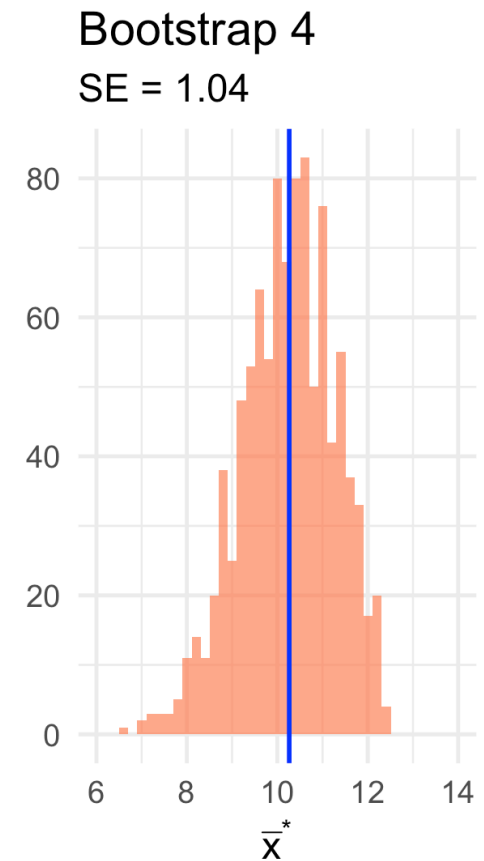
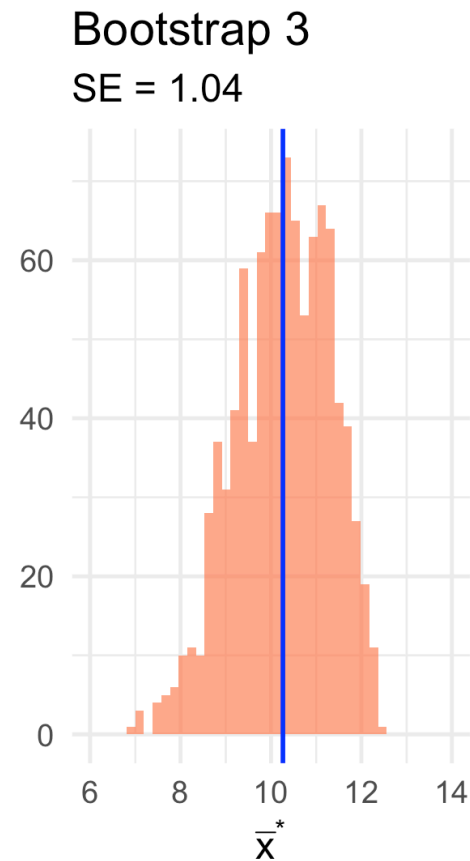
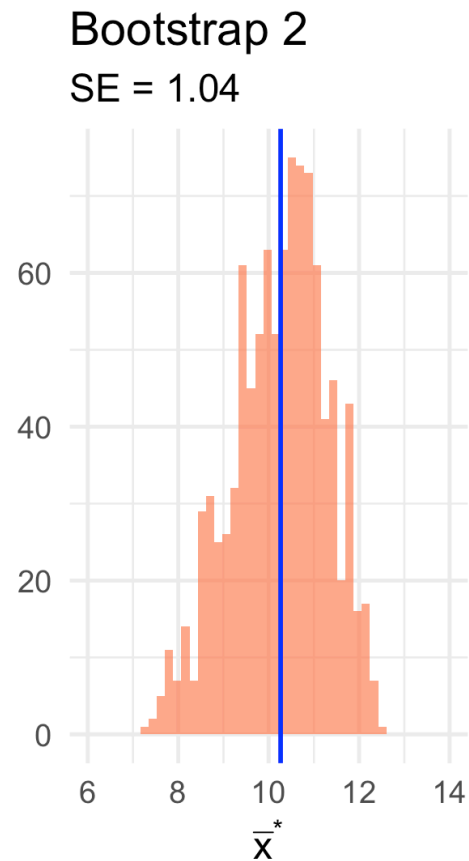
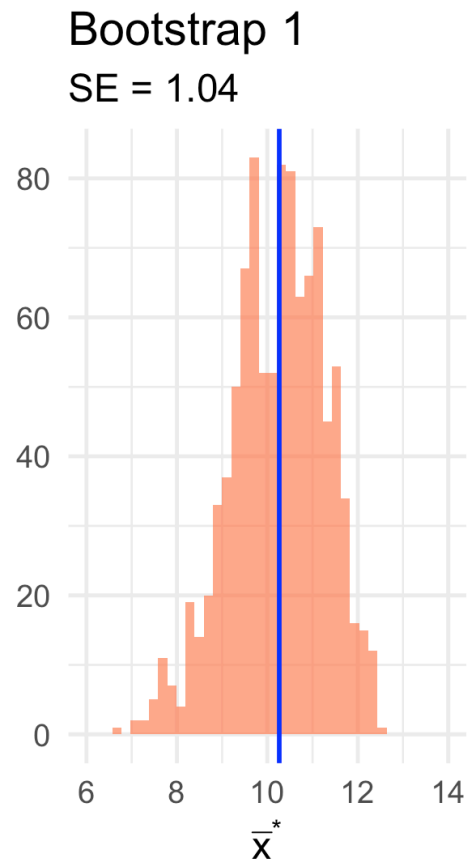
Bootstrap Failure Modes

The bootstrap is **not** a universal solution. It can fail when:

1. **Very small samples** ($n < 10$)
2. **Sample doesn't represent population**
3. **Estimating extreme quantiles**
4. **Discrete statistics** (e.g., median with few unique values)
5. **Heavy-tailed distributions** (infinite variance)
6. **Dependent data** (time series, spatial data)
7. **Lattice distributions** with small support

Failure Mode 1: Very Small Samples

► Code



Problem: Bootstrap distributions vary substantially due to small sample size.

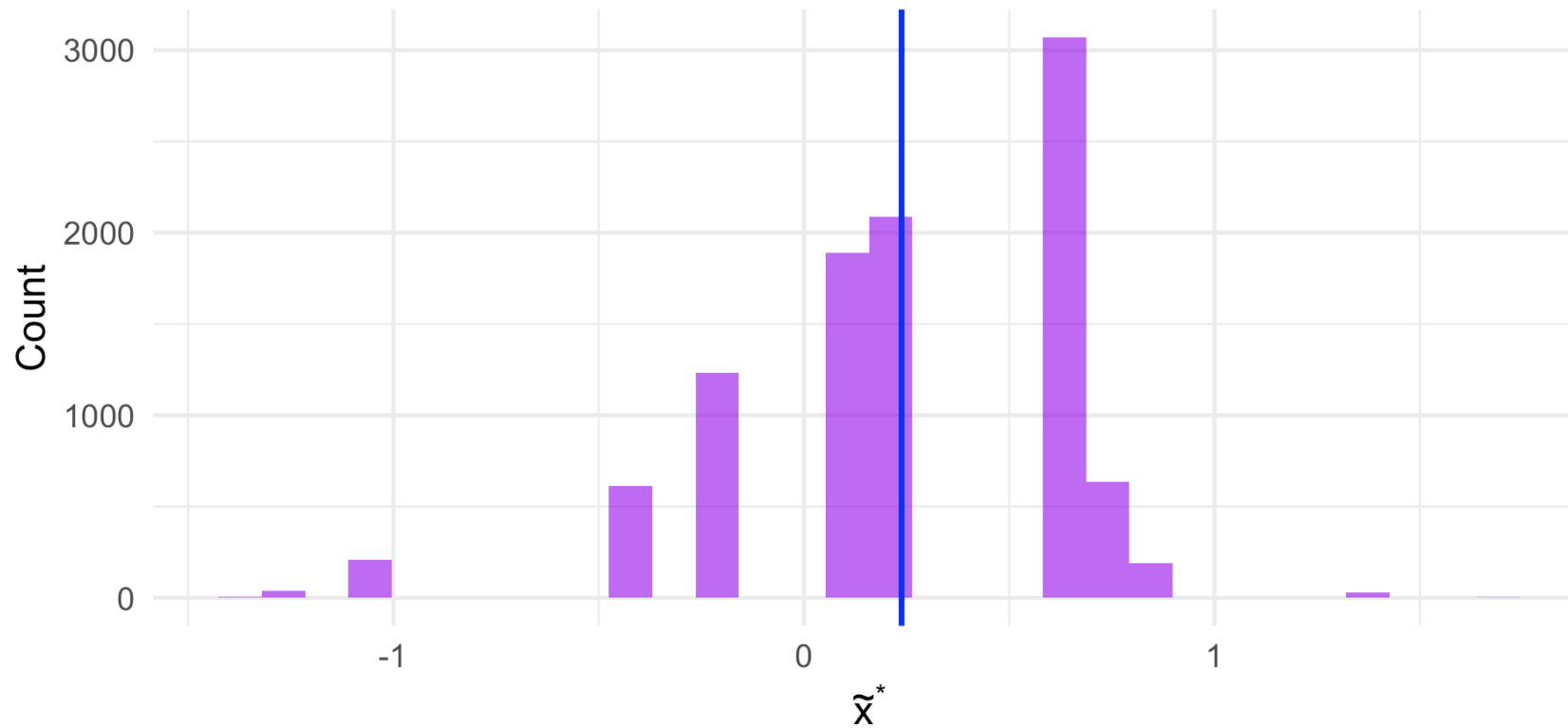


Failure Mode 2: The Median

► Code

Bootstrap Distribution of Median (n = 15)

Notice the discrete, irregular shape



Problem: Bootstrap medians only take values from the original sample!

Why Median Bootstrap Fails

The Median Problem (Chihara 5.8.3)

For odd sample size n , the bootstrap median can only take values that are **observations in the original sample**.

Consequences:

- Bootstrap distribution is **discrete**, not continuous
- Shape depends heavily on **gaps between observations**
- May poorly approximate the continuous sampling distribution

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Remedies:

- **Smoothed bootstrap:** Add small noise to observations
- **Larger samples:** Problem diminishes as n increases
- Use percentile intervals rather than bootstrap t

Failure Mode 3: Heavy-Tailed Distributions

! Cauchy Distribution Example

The Cauchy distribution has **no mean** (undefined). For such distributions:

- Sample mean is **not consistent**
- Bootstrap cannot “fix” an inconsistent estimator
- Bootstrap distribution doesn’t converge properly

```
1 # Cauchy has no mean – bootstrap can't help
2 set.seed(551)
3 cauchy_sample <- rcauchy(50)
4 boot_cauchy <- replicate(1000, mean(sample(cauchy_sample, 50, replace = TRUE)))
5
6 cat("Sample mean:", round(mean(cauchy_sample), 2), "\n")
```

Sample mean: 0.22

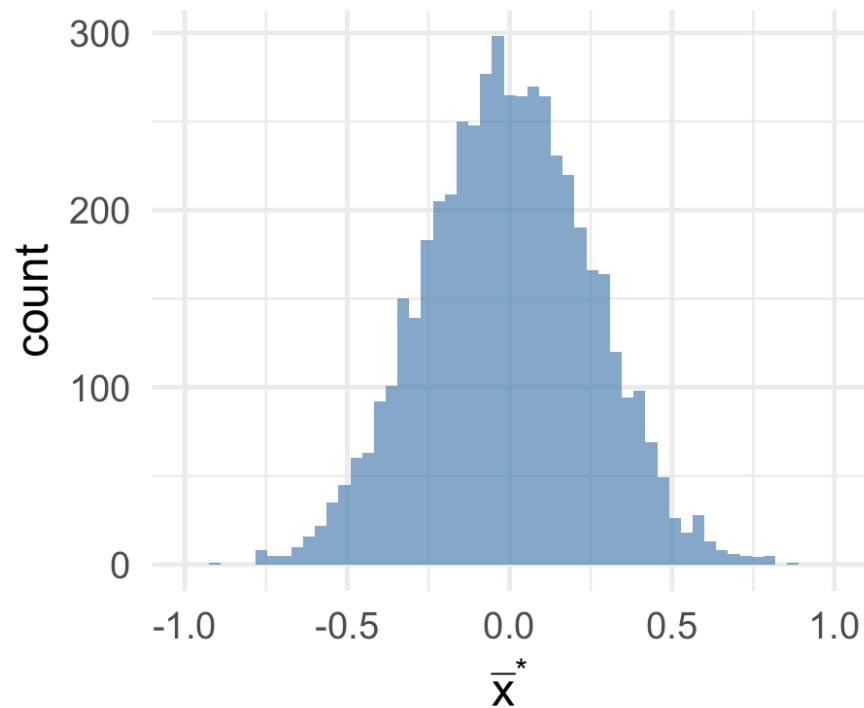
```
1 cat("Bootstrap SE:", round(sd(boot_cauchy), 2), " (meaningless!)"
```

Bootstrap SE: 0.57 (meaningless!)

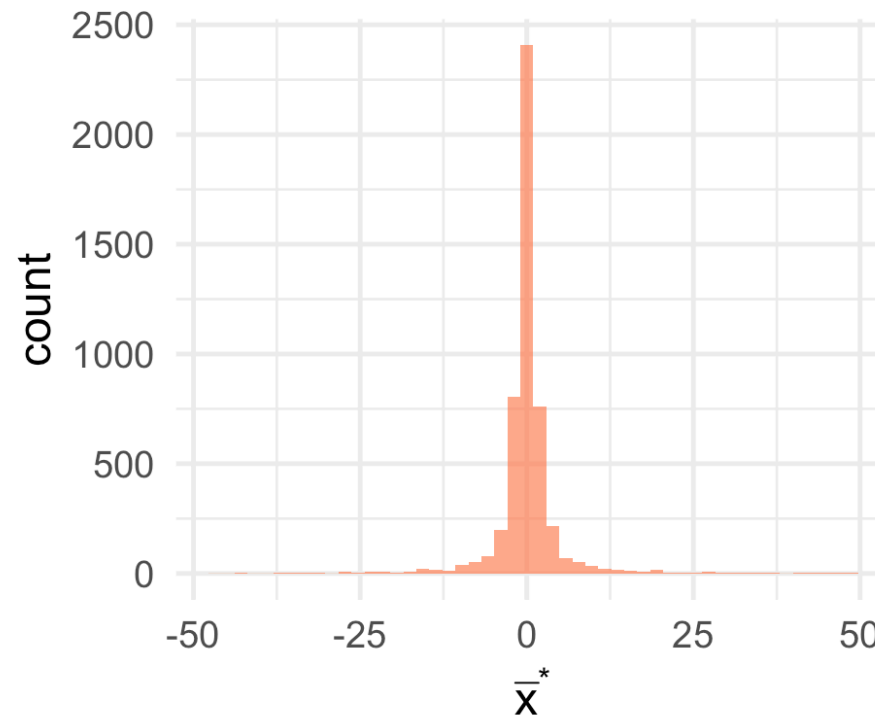
Visualizing Heavy-Tail Failure

► Code

Bootstrap Means: Normal(0,1)



Bootstrap Means: Cauchy (truncated)



For Cauchy, the bootstrap distribution is erratic and doesn't converge!

Failure Mode 4: Extreme Quantiles

⚠ Estimating Extreme Quantiles

Bootstrap has difficulty with extreme quantiles (e.g., 99th percentile) because:

1. Few observations near the tail
2. Bootstrap samples may not include extreme values
3. Standard bootstrap underestimates tail variability

```
1 # Bootstrap for 95th percentile
2 set.seed(551)
3 sample_data <- rexp(50, rate = 0.1)
4 boot_q95 <- replicate(5000, quantile(sample(sample_data, 50, replace = TRUE), 0.95))
5
6 cat("Sample 95th percentile:", round(quantile(sample_data, 0.95), 2), "\n")
```

Sample 95th percentile: 32.73

```
1 cat("Bootstrap SE:", round(sd(boot_q95), 2))
```

Bootstrap SE: 6.64

Failure Mode 4: Extreme Quantiles

⚠ Estimating Extreme Quantiles

Bootstrap has difficulty with extreme quantiles (e.g., 99th percentile) because:

1. Few observations near the tail
2. Bootstrap samples may not include extreme values
3. Standard bootstrap underestimates tail variability

```
1 # Bootstrap for 95th percentile
2 set.seed(551)
3 sample_data <- rexp(50, rate = 0.1)
4 boot_q95 <- replicate(5000, quantile(sample(sample_data, 50, replace = TRUE), 0.95))
5
6 cat("Sample 95th percentile:", round(quantile(sample_data, 0.95), 2), "\n")
```

Sample 95th percentile: 32.73

```
1 cat("Bootstrap SE:", round(sd(boot_q95), 2))
```

Bootstrap SE: 6.64

For extreme quantiles, consider **parametric bootstrap** or **extreme value theory**.

Summary: When Bootstrap Works and Fails

Works Well	Use Caution	May Fail
Sample mean	Median (small n)	Heavy tails (no variance)
Trimmed mean	Extreme quantiles	Dependent data
Correlation	Small samples ($n < 15$)	Non-representative sample
Regression coefficients	Heavily skewed data	Maximum/minimum
Smooth functions of means	Discrete distributions	Lattice distributions

Summary and Looking Ahead

Key Takeaways: The Bootstrap Principle

! The Bootstrap Principle

1. **The sample represents the population** — resample from the sample
2. **Sampling with replacement** mimics drawing new samples
3. **Bootstrap distribution approximates sampling distribution**
4. Use bootstrap to estimate **standard errors** and the **shape** of the sampling distribution

Key Takeaways: The Bootstrap Principle

! The Bootstrap Principle

1. **The sample represents the population** — resample from the sample
2. **Sampling with replacement** mimics drawing new samples
3. **Bootstrap distribution approximates sampling distribution**
4. Use bootstrap to estimate **standard errors** and the **shape** of the sampling distribution

Main advantages:

- No distributional assumptions required
- Works for **any statistic** (medians, ratios, etc.)
- Computationally straightforward

Why Resampling WITH Replacement?

Key Points

1. **Without replacement:** Would get the same sample every time (size n)
2. **With replacement:** Creates variability that mimics sampling from population
3. **Empirical distribution:** Assigns probability $1/n$ to each observation
4. **Number of bootstrap samples:** $\binom{2n-1}{n}$ possible — astronomically large!

Asymptotic Properties Summary

Bootstrap Consistency

Under regularity conditions:

- Bootstrap distribution **converges** to sampling distribution
- Bootstrap SE **consistently estimates** true SE
- Bootstrap t intervals have coverage error $O(n^{-1})$ – better than percentile!

Asymptotic Properties Summary

Bootstrap Consistency

Under regularity conditions:

- Bootstrap distribution **converges** to sampling distribution
- Bootstrap SE **consistently estimates** true SE
- Bootstrap t intervals have coverage error $O(n^{-1})$ — better than percentile!

The bootstrap accurately estimates:

- Spread (standard error)
- Shape (including skewness)
- Bias

But NOT the center (stays at $\hat{\theta}$, not θ)

When Bootstrap Fails: Summary

Failure Mode	Reason	Alternative
Very small n	Sample doesn't represent population	Classical methods
Median/quantiles	Discrete bootstrap distribution	Smoothed bootstrap
Heavy tails	Infinite variance	Subsampling
Extreme quantiles	Few observations in tails	Parametric bootstrap
Dependent data	Independence assumption violated	Block bootstrap

Looking Ahead

Next Lesson: Bootstrap Confidence Intervals

- Bootstrap standard errors
- Bootstrap percentile intervals
- Bootstrap t intervals
- Practical considerations and examples

Looking Ahead

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Coming Soon:

- Hypothesis testing framework
- Type I and Type II errors
- Power and sample size

References

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Questions?

Thank you!